

DATABASE MANAGEMENT SYSTEM LAB

ASSIGNMENT NO. 02 – [Name : Ridafatema Patel, Rollno. 63]

Consider the given relational table:

```
MariaDB [RIDA]> describe EMP;
```

Field	Type	Null	Key	Default	Extra
EmpNo	int(11)	NO	PRI	NULL	
EmpName	varchar(50)	YES		NULL	
Designation	varchar(50)	YES		NULL	
City	varchar(50)	YES		NULL	
Salary	int(11)	YES		NULL	
Zipcode	int(11)	YES		NULL	
Country	varchar(30)	YES		NULL	

7 rows in set (0.016 sec)

Q. Write SQL queries for the following :

1. Create a sequence used to generate employee numbers for the empno column of the emp table.

Before :

```
MariaDB [(none)]> use RIDA
Database changed
MariaDB [RIDA]> CREATE TABLE EMP (
-> EmpNo INT PRIMARY KEY,
-> EmpName VARCHAR(50),
-> Designation VARCHAR(50),
-> City VARCHAR(50),
-> Salary INT,
-> Zipcode INT,
-> Country VARCHAR(30)
-> );
```

After

```
MariaDB [RIDA]> ALTER TABLE EMP MODIFY EmpNo INT AUTO_INCREMENT;
```

Result :

```
MariaDB [RIDA]> select *from EMP;
```

EmpNo	EmpName	Designation	City	Salary	Zipcode	Country
101	Dnyaneshwar Pawar	Software Engineer	Mumbai	48000	71	India
102	Rida Patel	Data Analyst	Pune	55000	411	India
103	Aditi Kale	Data Analyst	Toronto	62000	881	Canada
104	Sneha Joshi	HR Executive	London	55000	541	UK
105	Satyam Singh	Project Manager	New York	90000	100	USA
106	Om Dhole	UI Designer	Sydney	46000	200	Australia
107	Gaurav Pagare	Backend Developer	San Francisco	70000	10115	USA
108	Pradip Jire	AI Engineer	Nagpur	95000	941	Germany
109	Kashish Sonawane	Tester	Nagpur	30000	440	India
110	Aarati Kale	Frontend Developer	Vancouver	60000	12345	Canada
111	Prerana Mahajan	Backend Engineer	Parbhani	78903	71	India

2. Create an Index on Country.

```
MariaDB [RIDA]> CREATE INDEX idx ON EMP(Country);
```

3. Find the Country whose zipcode = 071 and check whether the query uses the Index and write your observation.

```
MariaDB [RIDA]> SELECT EmpNo, EmpName, Country FROM EMP WHERE ZipCode=71;
+-----+-----+-----+
| EmpNo | EmpName          | Country |
+-----+-----+-----+
| 101   | Dnyaneshwar Pawar | India   |
| 111   | Prerana Mahajan   | India   |
+-----+-----+-----+
```

```
MariaDB [RIDA]> SHOW INDEX FROM EMP;
```

Table	Non unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment
emp	0	PRIMARY	1	EmpNo	A	11	NULL	NULL		BTREE		
emp	1	idx	1	Country	A	11	NULL	NULL	YES	BTREE		

```
MariaDB [RIDA]> EXPLAIN SELECT COUNTRY FROM EMP WHERE ZipCode =71;
```

id	select_type	table	type	possible_keys	key	key_len	ref	rows	Extra
1	SIMPLE	EMP	ALL	NULL	NULL	NULL	NULL	11	Using where

4. Create a view for Employees having salary<50000 and stays in 'Mumbai'.

```
MariaDB [RIDA]> CREATE VIEW v1 AS SELECT EmpNo, EmpName, Designation, Country FROM EMP;
```

```
MariaDB [RIDA]> SELECT * FROM v1;
```

EmpNo	EmpName	Designation	Country
101	Dnyaneshwar Pawar	Software Engineer	India
102	Rida Patel	Data Analyst	India
103	Aditi Kale	Data Analyst	Canada
104	Sneha Joshi	HR Executive	UK
105	Satyam Singh	Project Manager	USA
106	Om Dhole	UI Designer	Australia
107	Gaurav Pagare	Backend Developer	USA
108	Pradip Jire	AI Engineer	Germany
109	Kashish Sonawane	Tester	India
110	Aarati Kale	Frontend Developer	Canada
111	Prerana Mahajan	Backend Engineer	India

5. Display count of Employees who stays in Mumbai using View.

```
MariaDB [RIDA]> CREATE VIEW v2 AS SELECT City, Country, COUNT(City) AS c FROM EMP WHERE City = 'Mumbai';
Query OK, 0 rows affected (0.007 sec)
```

```
MariaDB [RIDA]> SELECT * FROM v2;
```

City	Country	c
Mumbai	India	1

6. Find the average salary of employees from a created view.

```
MariaDB [RIDA]> CREATE VIEW v3 AS SELECT EmpNo, EmpName, Salary, City FROM EMP;
```

```
MariaDB [RIDA]> SELECT AVG(Salary) AS avgg FROM v3;
```

avgg
62718.4545

7. Display EmpNames who stays on the same street (consider city)of a view.

```
MariaDB [RIDA]> SELECT EmpName FROM v3 WHERE City = 'Nagpur';
```

EmpName
Pradip Jire
Kashish Sonawane